
2019-Ph-H1-Q4

Section: Particles and Waves

Topic: Radiation

Question:

Which radiation is most ionising?

- A) Alpha
- B) Beta
- C) Gamma
- D) X-rays
- E) Neutrons

Answer:

Alpha radiation is the most ionising because of its **high charge and mass**, despite low penetration.

Correct Option: A

Guidance for Students:

Ionising power is about how much damage particles can cause to atoms by knocking out electrons.

Revision Tips:

- Alpha: heavy, slow, **highly ionising**, blocked by paper
- Beta: lighter, **moderately ionising**, blocked by aluminium
- Gamma: waves, **weakly ionising**, needs lead to block
- More ionising = **more dangerous** inside the body

